

Programming Lab 2 Assignment

P.Balamurugan, Urban Larsson, Sahil Wagh, Nitish Dumoliya

August 7, 2024

1. Matrix Transposition

- **Problem:** Write a program to transpose a given 2D matrix. The transpose of a matrix is obtained by swapping its rows with columns.
- **Input:** A 2D matrix M of size $m \times n$.
- **Output:** The transposed matrix of size $n \times m$.
- **Example:**

```
Matrix M:  
[[1, 2, 3],  
 [4, 5, 6]]
```

```
Transposed Matrix:  
[[1, 4],  
 [2, 5],  
 [3, 6]]
```

2. Matrix Row and Column Sum

- **Problem:** Given a 2D matrix, calculate the sum of each row and each column, and store these sums in dictionaries.
- **Input:** A 2D matrix M of size $m \times n$.
- **Output:** Two dictionaries:
 - One dictionary with row indices as keys and row sums as values.
 - Another dictionary with column indices as keys and column sums as values.
- **Example:**

```
Matrix M:  
[[1, 2, 3],  
 [4, 5, 6],  
 [7, 8, 9]]
```

```
Row Sums: {0: 6, 1: 15, 2: 24}  
Column Sums: {0: 12, 1: 15, 2: 18}
```

3. Matrix Diagonal Elements Dictionary

- **Problem:** Create a dictionary to store the diagonal elements of a square matrix. The keys should be "primary" and "secondary," and the values should be lists of diagonal elements.
- **Input:** A square matrix M of size $n \times n$.
- **Output:** A dictionary with primary and secondary diagonal elements.
- **Example:**

Matrix M:
[[1, 2, 3],
 [4, 5, 6],
 [7, 8, 9]]

Diagonals: {'primary': [1, 5, 9], 'secondary': [3, 5, 7]}

4. Matrix Flattening and Frequency Count

- **Problem:** Flatten a 2D matrix into a 1D list and then use a dictionary to count the frequency of each element in the flattened list.
- **Input:** A 2D matrix M of any size.
- **Output:** A dictionary with matrix elements as keys and their frequencies as values.
- **Example:**

Matrix M:
[[1, 2, 2],
 [3, 1, 4],
 [2, 3, 3]]

Flattened List: [1, 2, 2, 3, 1, 4, 2, 3, 3]

Frequency Count: {1: 2, 2: 3, 3: 3, 4: 1}

5. Matrix Elements Frequency in Rows and Columns

- **Problem:** Create a dictionary where each key is an element of the matrix, and the value is another dictionary containing the frequency of the element in each row and each column.
- **Input:** A 2D matrix M of size $m \times n$.
- **Output:** A dictionary with elements as keys, and each value is a dictionary with 'rows' and 'columns' as keys, containing frequency counts.
- **Example:**

Matrix M:
[[1, 2, 1],
 [1, 3, 2],
 [3, 1, 1]]

Element Frequency:
{1: {'rows': [2, 1, 3], 'columns': [3, 2, 1]},
 2: {'rows': [1, 2, 0], 'columns': [1, 1, 0]},
 3: {'rows': [0, 1, 1], 'columns': [1, 1, 1]}}